COL724/7524 Advanced Computer Networks
Assignment 1 – Part A
Baseline Delay Calculation

1. Network Path

The baseline topology consists of four autonomous systems connected in a linear path:

$$\text{AS1} \rightarrow \text{AS2} \rightarrow \text{AS3} \rightarrow \text{AS4}$$

The propagation delays of the three links are:

Link	**Propagation Delay**
AS1–AS2	3 ms
AS2–AS3	4 ms
AS3–AS4	3 ms

2. Theoretical Minimum Propagation Delay

The theoretical minimum one-way propagation delay is obtained by summing the propagation delays of all three links:

$$D_{\text{prop}} = D_{\text{AS1}-\text{AS2}} + D_{\text{AS2}-\text{AS3}} + D_{\text{AS3}-\text{AS4}}$$

$$D_{\text{prop}} = 3 + 4 + 3$$

$$\boxed{D_{\text{prop}} = 10 \text{ ms}}$$

This value represents only the propagation delay and does not include packet transmission time or other processing effects.

3. Transmission Delay

The packets generated by the application have a packet size of 540 bytes according to the FlowMonitor output.

The transmission delay on a link is:

$$D_{\text{tx}} = \frac{L}{R}$$

where L is the packet size in bits and R is the link data rate.
The packet size is:

$$L = 540 \times 8 = 4320 \text{ bits}$$

For the AS1–AS2 link with rate 8 Mbps:

$$D_{\text{tx},1} = \frac{4320}{8 \times 10^6} = 0.54 \text{ ms}$$

For the AS2–AS3 link with rate 6 Mbps:

$$D_{\text{tx},2} = \frac{4320}{6 \times 10^6} = 0.72 \text{ ms}$$

For the AS3–AS4 link with rate 8 Mbps:

$$D_{\text{tx},3} = \frac{4320}{8 \times 10^6} = 0.54 \text{ ms}$$

Therefore, the total transmission delay is:

$$D_{\text{tx,total}} = 0.54 + 0.72 + 0.54$$

$$\boxed{D_{\text{tx,total}} = 1.80 \text{ ms}}$$

4. Expected End-to-End Delay

Including both propagation and transmission delays:

$$D_{\text{expected}} = D_{\text{prop}} + D_{\text{tx,total}}$$

$$D_{\text{expected}} = 10 + 1.80$$

$$\boxed{D_{\text{expected}} = 11.80 \text{ ms}}$$

5. FlowMonitor Average Delay

The FlowMonitor XML reports:

$$\text{delaySum} = 9.44533 \times 10^7 \text{ ns}$$

and

$$\text{rxPackets} = 8$$

The average delay is calculated as:

$$D_{\text{avg}} = \frac{\text{delaySum}}{\text{rxPackets}}$$

Therefore,

$$D_{\text{avg}} = \frac{9.44533 \times 10^7}{8}$$

$$D_{\text{avg}} = 1.18066625 \times 10^7 \text{ ns}$$

Converting nanoseconds to milliseconds:

$$D_{\text{avg}} = \frac{1.18066625 \times 10^7}{10^6}$$

$$\boxed{D_{\text{avg}} \approx 11.8067 \text{ ms}}$$

6. Comparison

The theoretical propagation-only delay is:

$$\boxed{10.00 \text{ ms}}$$

The expected delay after including transmission time is:

$$\boxed{11.80 \text{ ms}}$$

The FlowMonitor measured average delay is:

$$\boxed{11.8067 \text{ ms}}$$

The difference between the propagation-only delay and the measured average is:

$$11.8067 - 10.00 = 1.8067 \text{ ms}$$

This difference is primarily due to packet transmission time on the three links. The calculated propagation plus transmission delay of approximately 11.80 ms is very close to the FlowMonitor value of 11.8067 ms. The small remaining difference is due to simulation and protocol processing effects.

7. Packet Delivery

The FlowMonitor output reports:

$$\text{txPackets} = 8, \qquad \text{rxPackets} = 8, \qquad \text{lostPackets} = 0$$

Thus, all transmitted packets were successfully delivered:

$$\text{Delivery Rate} = \frac{8}{8} \times 100 = \boxed{100\%}$$

and the packet loss was:

$$\boxed{0\%}$$

8. Console Output

The following screenshot shows the console output obtained when running the baseline `bgp.cc` simulation.

```
soham-malvadar@Malvadar-OMEN:~/ns-3.48$ ./ns3 run scratch/bgp.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
Starting Inter-AS routing simulation with 4 routers (AS1-AS2-AS3-AS4)...
At time +2s client sent 512 bytes to 192.168.3.2 port 8080
At time +2.01181s server received 512 bytes from 192.168.1.1 port 49153
At time +2.01181s server sent 512 bytes to 192.168.1.1 port 49153
At time +2.02361s client received 512 bytes from 192.168.3.2 port 8080
At time +2.75s client sent 512 bytes to 192.168.3.2 port 8080
At time +2.76181s server received 512 bytes from 192.168.1.1 port 49153
At time +2.76181s server sent 512 bytes to 192.168.1.1 port 49153
At time +2.77361s client received 512 bytes from 192.168.3.2 port 8080
At time +3.5s client sent 512 bytes to 192.168.3.2 port 8080
At time +3.51181s server received 512 bytes from 192.168.1.1 port 49153
At time +3.51181s server sent 512 bytes to 192.168.1.1 port 49153
At time +3.52361s client received 512 bytes from 192.168.3.2 port 8080
At time +4.25s client sent 512 bytes to 192.168.3.2 port 8080
At time +4.26181s server received 512 bytes from 192.168.1.1 port 49153
At time +4.26181s server sent 512 bytes to 192.168.1.1 port 49153
At time +4.27361s client received 512 bytes from 192.168.3.2 port 8080
At time +5s client sent 512 bytes to 192.168.3.2 port 8080
At time +5.01181s server received 512 bytes from 192.168.1.1 port 49153
At time +5.01181s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.02361s client received 512 bytes from 192.168.3.2 port 8080
At time +5.75s client sent 512 bytes to 192.168.3.2 port 8080
At time +5.76181s server received 512 bytes from 192.168.1.1 port 49153
At time +5.76181s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.77361s client received 512 bytes from 192.168.3.2 port 8080
At time +6.5s client sent 512 bytes to 192.168.3.2 port 8080
At time +6.51181s server received 512 bytes from 192.168.1.1 port 49153
At time +6.51181s server sent 512 bytes to 192.168.1.1 port 49153
At time +6.52361s client received 512 bytes from 192.168.3.2 port 8080
At time +7.25s client sent 512 bytes to 192.168.3.2 port 8080
At time +7.26181s server received 512 bytes from 192.168.1.1 port 49153
At time +7.26181s server sent 512 bytes to 192.168.1.1 port 49153
At time +7.27361s client received 512 bytes from 192.168.3.2 port 8080
soham-malvadar@Malvadar-OMEN:~/ns-3.48$
```

Figure 1: Console output of the baseline simulation.